

Ziyi Yang (杨子毅)

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Expected Graduation: 2028-06

Date of Birth.: 2001-04



Education

Ph.D.	Organic Chemistry	Tsinghua University	2023 – Now
B.S.	Chemical Biology	Tsinghua University	2019 – 2023

Research: Computational Protein Design & Chemical Protein Synthesis

1) Design Cyclic Peptide Binders

Built **million-scale** cyclic-peptide–receptor structure datasets by mimicking intra-protein interfaces and screening cyclic-compatible conformations (CPSea1: residue distance; CPSea2: terminal amide pose), enabling model scaling.

Z. Yang, H. Xie, Y. Jia, et al. *NeurIPS*, **2025**. <https://openreview.net/pdf?id=CfOsKx3jF8>

Z. Yang[†], Y. Jia[†], et al. *under review*, **2026**. <https://huggingface.co/datasets/YZY010418/CPSea2>

2) Design Mirror-Image Peptide Binders

Introduced axial-vector / pseudo-scalar features to **explicitly encode molecular chirality**, enabling generative models to match training-set chirality, generalize across chirality for hetero-chiral design, and control chirality patterns.

Z. Yang[†], Z. Tian[†], Y. Jia[†], et al. *ICML*, **2026**. <https://arxiv.org/abs/2602.20176>

3) Assist Protein Folding by Glycosylation

Showed glycosylation prevents aggregation during folding, enabling synthesis of multiple proteins previously considered hard to fold.

T. Wang[†], W. Shi[†], **Z. Yang**, et al. *Angew. Chem.*, **2024**, 136, e202313640.

4) Triple-Split Inteins for Total Synthesis of Ultra-Large Proteins

Discovered a split intein retaining high splicing efficiency when split into three segments, overcoming concentration / denaturant limits, achieved **total synthesis of a >100 kDa protein**.
Manuscript in preparation

Visit and Intern

2026.04 – Now <i>Peptide drug design</i>	Bytedance AI drug discovery	Intern
2025.03 – Now <i>Protein design (synthetic data and generative models)</i>	Yanyan Lan Lab (AIR, Tsinghua)	Intern
2023.06 – 2023.07 <i>Direct evolution of split inteins via phage display</i>	Jishen Zheng Lab (USTC)	Summer Visit
2022.07 – 2023.08 <i>Design mirror-image peptide binders via the RIF pipeline</i>	Peilong Lu Lab (Westlake Univ.)	Summer Visit